SHUBHANKAR MILIND POTDAR

smpotdar@alumni.cmu.edu | +16043492981 | Papers and demos: smpotdar.github.io

EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, PA **May 2020**
Masters in Electrical and Computer Engineering.
Graduate Courses: Robot localization and mapping, Intro to ML, Computer Vision, Intermediate Deep Learning, ML for Structured Data, Advanced Probability and Statistics, Foundations of Computer Systems, Distributed Systems.

Veermata Jijabai Technological Institute (VJTI), University of Mumbai, India **May 2016**
Bachelor of Technology in Electrical Engineering.

PUBLICATIONS

- A Harley, Y Zuo, J Wen, A Mangal, **S. Potdar**, R. Chaudhry and K Fragkiadaki. "*Track, Check, Repeat: An EM Approach to Unsupervised Tracking.*" **CVPR 2021.** Paper Page
- M Prabhudesai*, S Lal*, HYF Tung, A Harley, **S. Potdar**, and K Fragkiadaki. "*3DQ-Nets: Visual Concepts Emerge in Pose Equivariant 3D Quantized Neural Scene Representations.*" **CVPR Workshops 2020.** Paper Page
- **S. Potdar**, S. Pund, S. Shende, S.Lote, K. Kanakgiri and F. Kazi. "*Real-time localisation and path-planning in ackermann steering robot using a single RGB camera and 2D LIDAR*". **IEEE ICIIECS 2017** Paper Page
- **S. Potdar**, A. Sawarkar, and F. Kazi. "*Learning from demonstration from multiple agents in humanoid robot.*" **IEEE SCEECS 2016** Paper Page

RELEVANT PROFESSIONAL EXPERIENCE

Organization: Koh Young Research Canada(KYRC), Vancouver, BC, Canada
KYRC is the research arm of Koh Young Technology(KYT), which is a company that builds machine vision technologies for providing optical inspection and measurement solutions for chip and PCB manufacturing factories.
Position: R&D Engineer **August 2021 - Present**
- Responsible for designing, implementing and testing algorithm prototypes from scratch in Python, and deploying in C++, using a mixture of CV, ML and deep learning techniques to achieve almost 99% accuracy on test datasets and live inspection conditions at customer locations.
- Project - Flux AI - Segmentation: Developed a deep learning based segmentation system to learn and rapidly identify regions of PCB which are covered with flux. Flux is a fluid applied on the PCB , before soldering components like chips and connectors to the PCB. Achieved perfect segmentation for customers like Qualcomm and Huawei in test conditions.
- Project - Flux AI - Classification: Designed and created an image classification model training tool that allows users without a background in CV/ML to train image classifiers, which classify if a component has flux or not, at customer site.
- Project - Pad detection algorithm: Developed an algorithm which rapidly detects all the solder pads in a RGB image of a Printed Circuit Board (PCB) using only unsupervised machine learning and template matching.

Organization: Machine Learning Department, Carnegie Mellon University, PA, USA
Position: Research Intern - Katerina Fragkiadaki Lab **July 2020 - April 2021**
- Responsible for implementing models from research papers using pytorch, conducting experiments and contributing to new research papers.
- Project - Track, Check, Repeat: Contributed in developing an EM based unsupervised technique to track 3D objects in videos and reported state-of-the-art accuracy in object discovery and tracking on CATER and KITTI. Paper Page
- Project - 3DQ-Net: Contributed in developing a model to detect objects in 3D without 3D supervision and outperform baselines . Paper Page

Organization: Cylab, Carnegie Mellon University, Pittsburgh, PA, USA
Cylab is a lab in Carnegie Mellon that focuses on cyber security and computer vision problems.
Position: Summer Research Assistant **May 2019 - August 2019**
- Responsible for conducting research and developing deep learning based models for the problems proposed by the lab's industrial partners.
- Project - Product Image Classification: Developed ResNet and VGGNet based models for BossaNova robotics to perform classification of the products seen by the robots moving in the aisle. Solved the problem of imbalanced dataset by using various losses like focal-loss, ring-loss and techniques like imbalanced sampling to achieve test accuracy of 97%.

SKILLS AND PROGRAMMING LANGUAGES

- Programming Languages: Python (Level: Advance), C++(Level: Intermediate), C (Level: Beginner), Java(Level: Beginner)
- Software Libraries: PyTorch, TensorFLow, OpenCV.

EXTRA CURRICULAR ACTIVITIES, COMPETITIONS AND AWARDS

- Runner up at the AngelHack hackathon for Augmented Reality Storytelling.
- Awarded Intel RealSense 3D camera by Intel Corporation for being a finalist Intel RealSense App challenge.